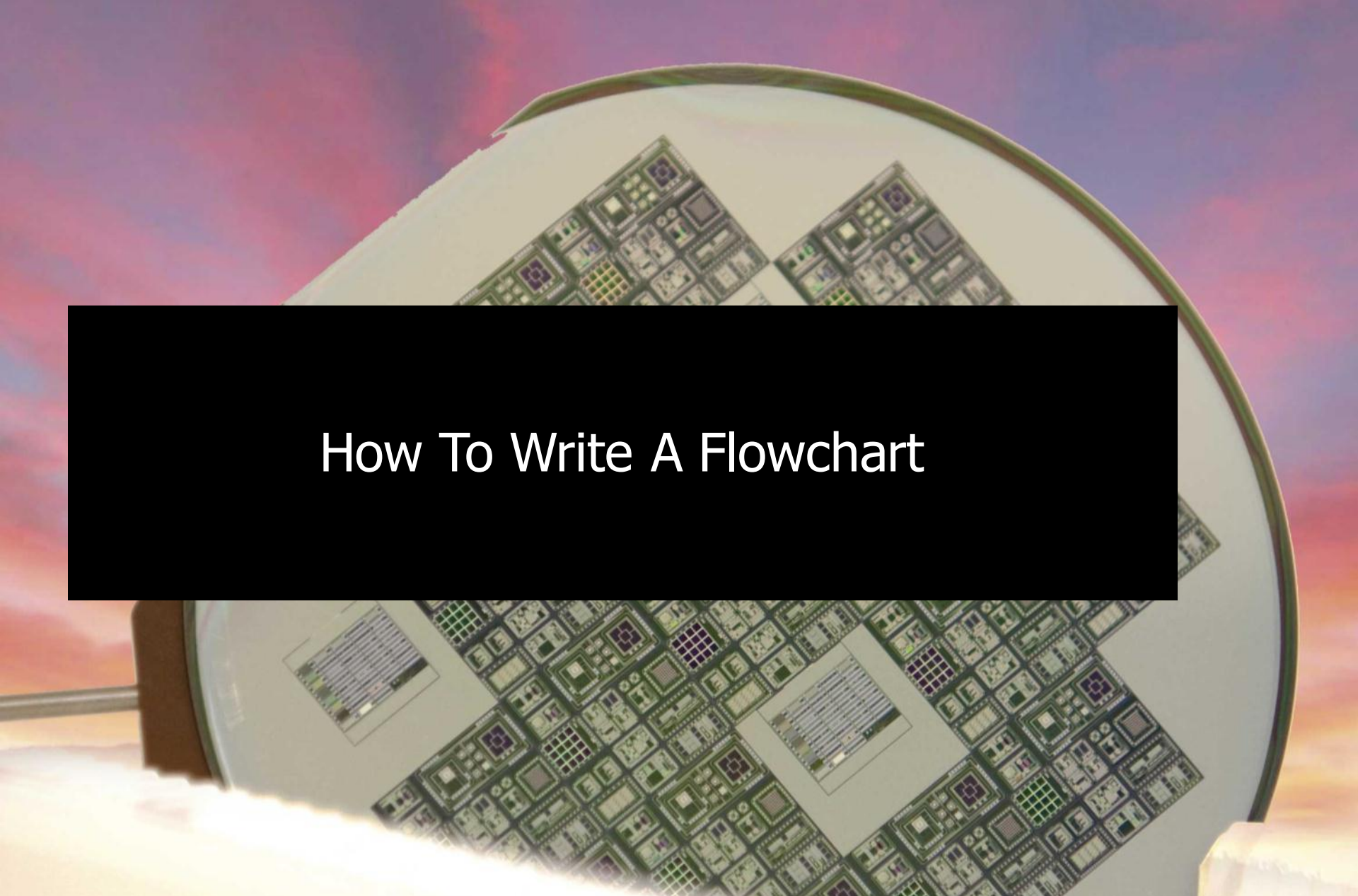




How To Write A Flowchart

Learning Objectives

- To learn how you transfer a device concept into a process flow to fabricate the device in the EKL labs
- You learn the different components that makes up a flowchart; process blocks, process modules and process steps
- you learn to match a concept process flow to process blocks, modules and steps
-



Course Content

1. Introduction
2. Flowchart elements: process blocks, modules and steps
3. Making a flowchart



How to write a flowchart

1. Introduction
2. Flowchart elements: process blocks, modules and steps
3. Making a flowchart



How to write a flowchart

1. Introduction

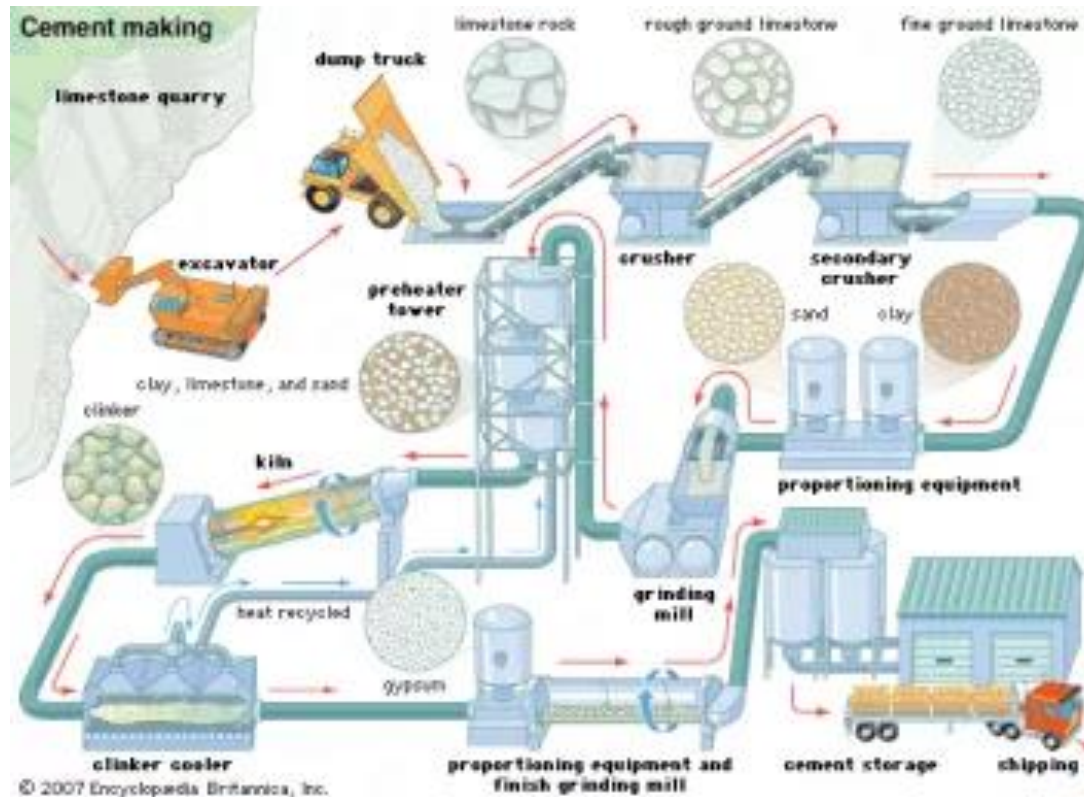
2. Flowchart elements: process blocks, modules and steps

3. Making a flowchart



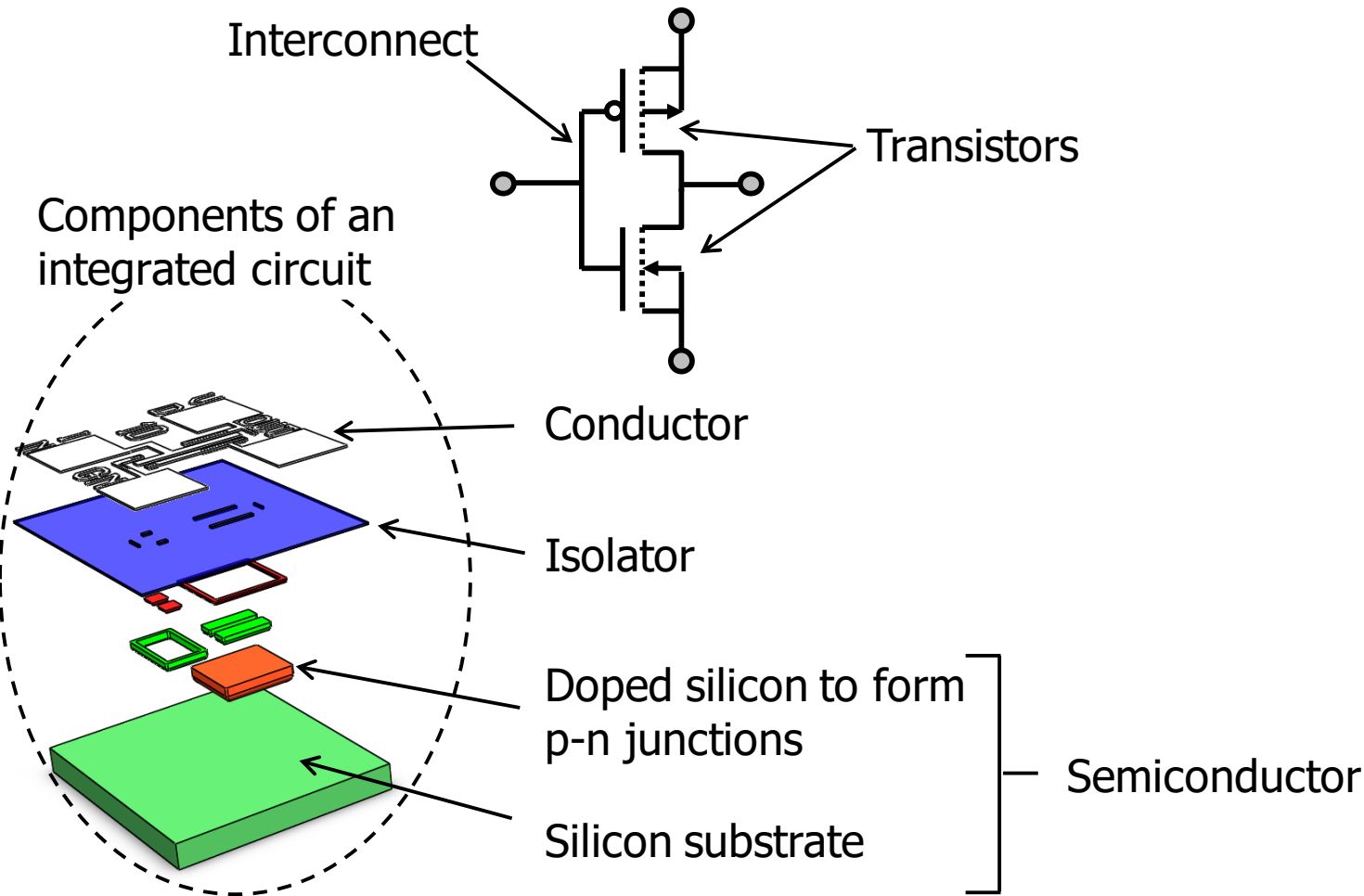
Flowcharts: An industrial approach to specify a production process

Cement Manufacturing Process Simplified Flow Chart

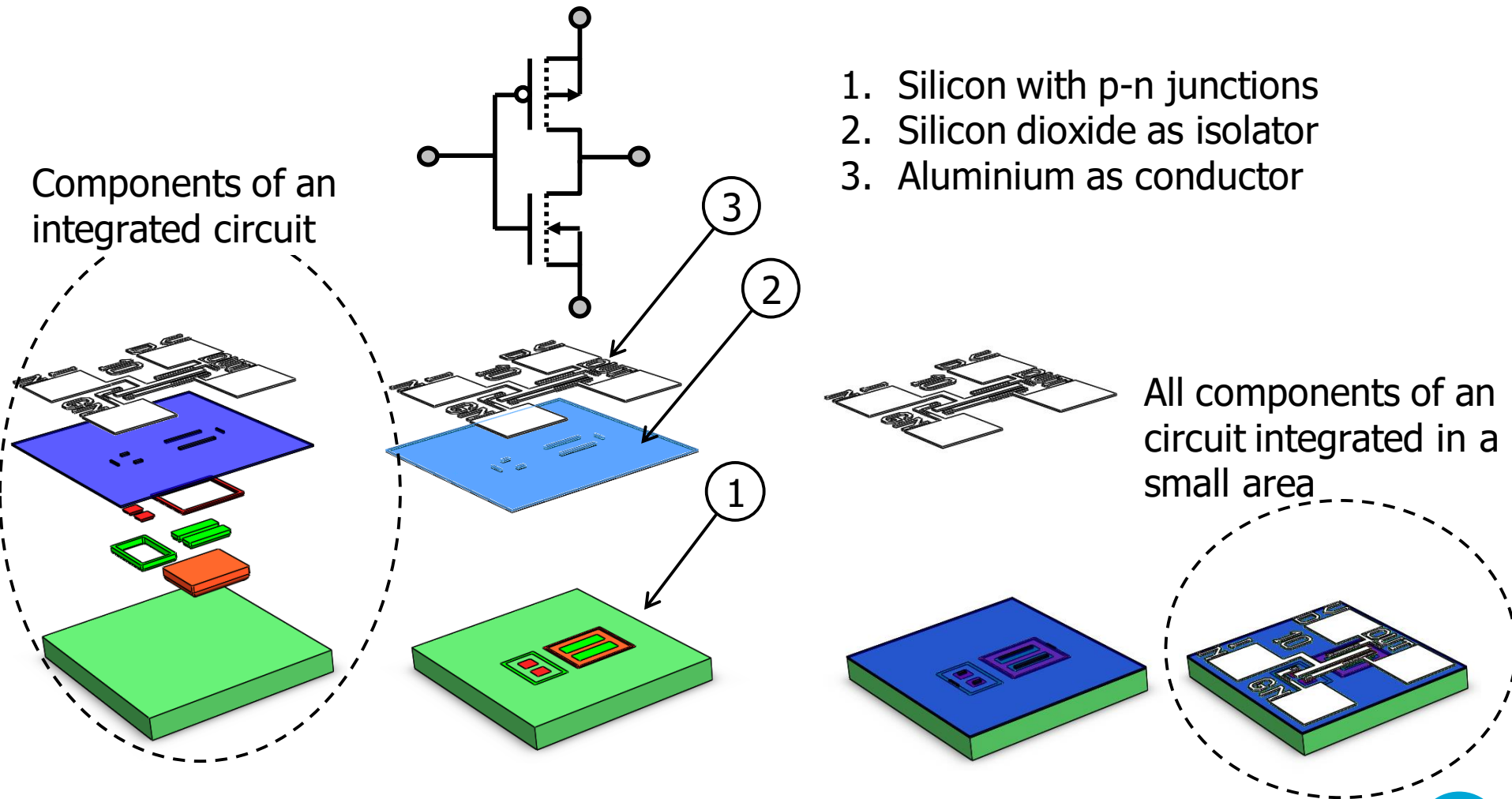


Source: <http://civildigital.com/cement-manufacturing-process-simplified-flow-chart/>

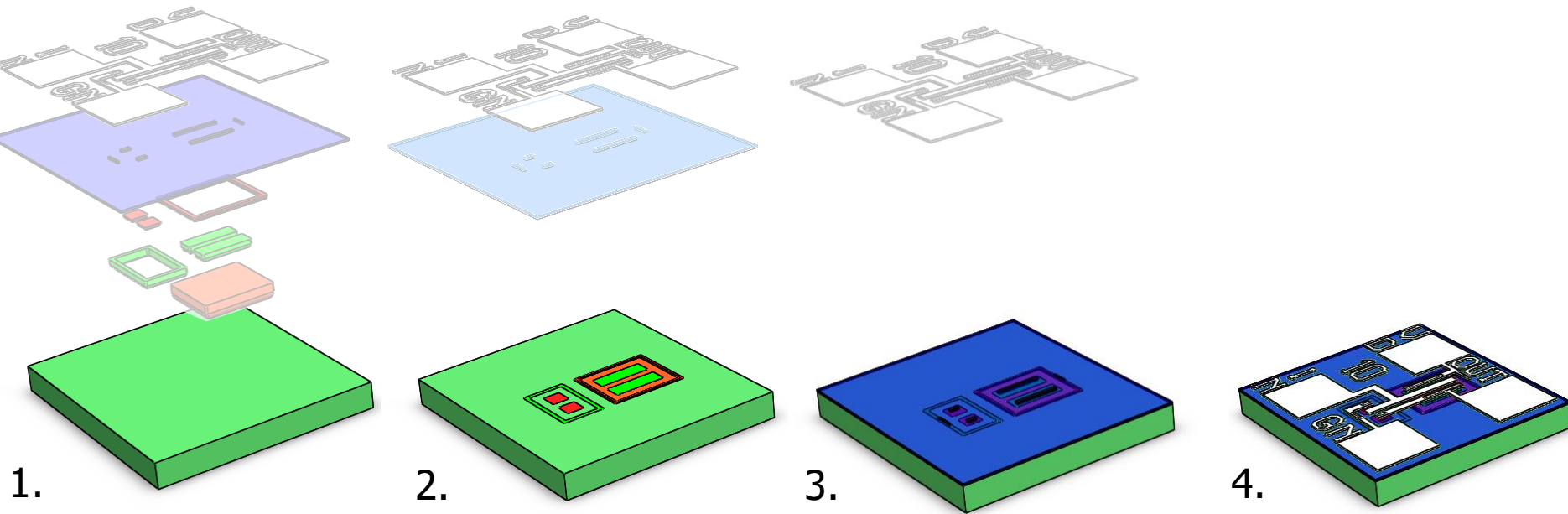
How can we turn a piece of semiconductor material into an integrated circuit?



How can we turn a piece of silicon into an integrated circuit?



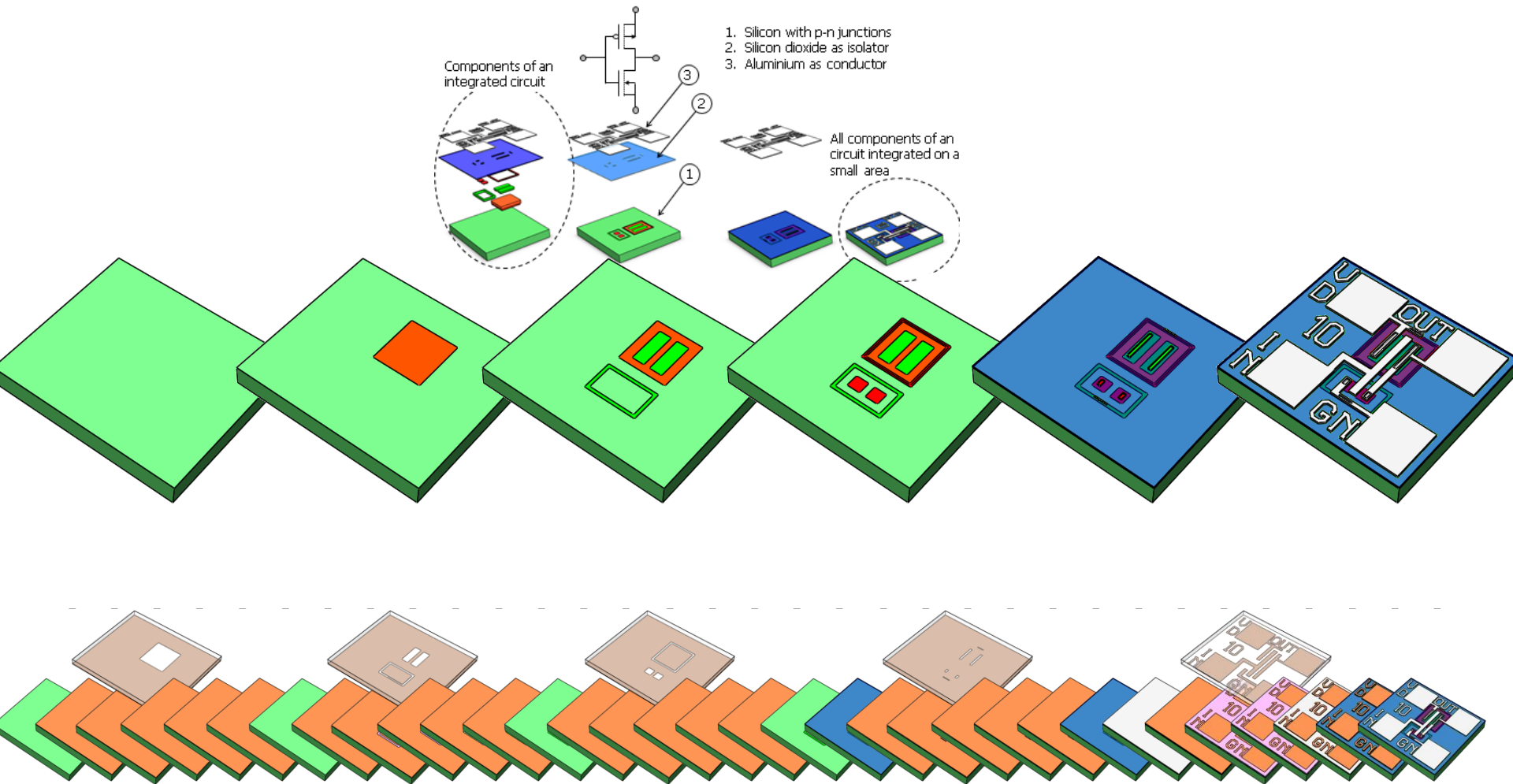
How can we turn a piece of silicon into an integrated circuit?



1. Start with a pure semiconductor.
2. Add different kind of atoms through the **SURFACE** into the semiconductor to change the electronic properties in specific **PATTERNS**.
3. Isolate the **SURFACE** of the semiconductor with an insulating **THIN FILM**.
4. Apply a **THIN FILM** of metal on the **SURFACE, PATTERN** this film to connect the different areas of the semiconductor.

How can we turn a piece of silicon into an integrated circuit?

How can we turn a piece of semiconductor material into an integrated circuit?

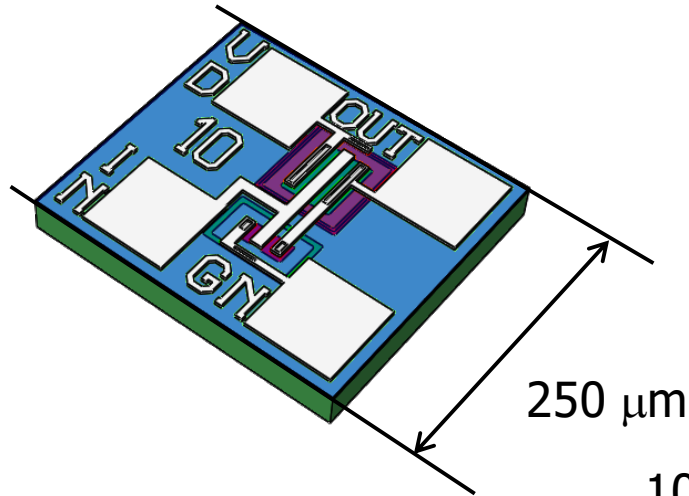


IC fabrication characteristics; 1 - planar technology

Thin film deposition	Oxidation of silicon	SURFACE $\text{Si} + \text{O}_2 \rightarrow \text{SiO}_2$
	Deposition	SURFACE reactions from the gas phase LPCVD, PECVD or sputtering (PVD)
	Electro plating	Electro chemical deposition on the SURFACE
Doping	Ion implantation	Ions implanted via the SURFACE
	Diffusion	Dopant diffusion via the SURFACE
Lithography	Coat photoresist	Apply a thin photosensitive film on the SURFACE
	Align and Expose	Align on markers on the SURFACE and expose the SURFACE
	Develop	Dissolve exposed photoresist from the SURFACE
Etch	wet	Dissolve reaction products from the SURFACE
	dry	Evaporate reaction products from the SURFACE

IC fabrication characteristics; 1 - planar technology

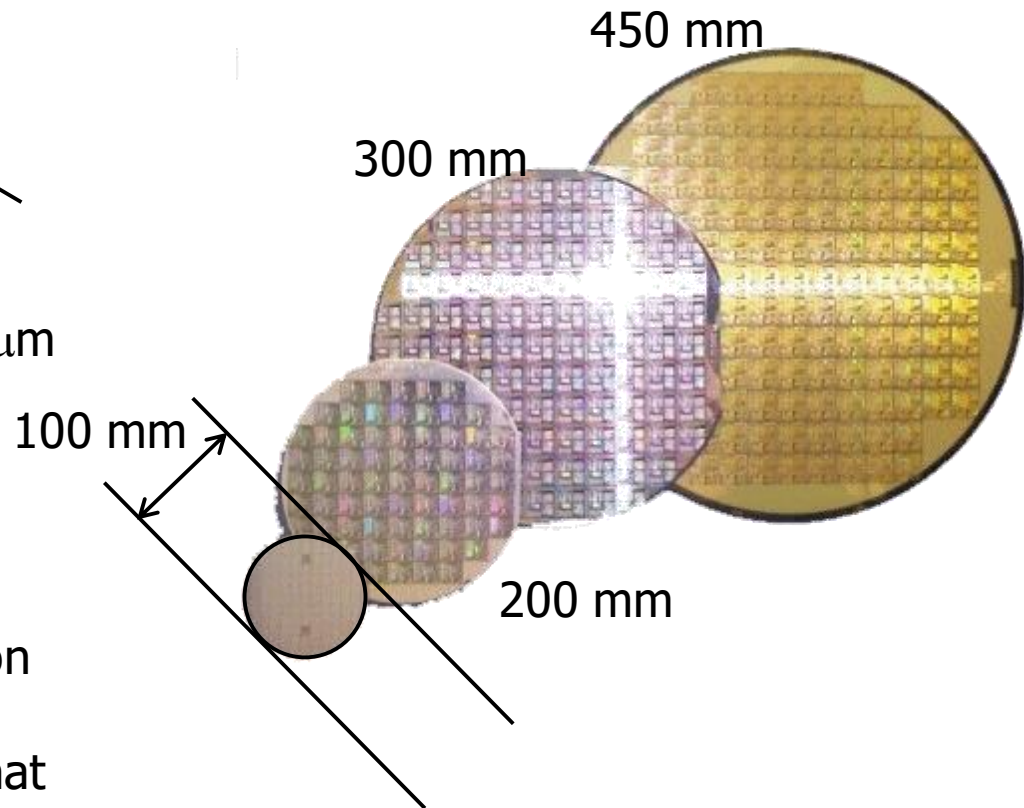
Device area



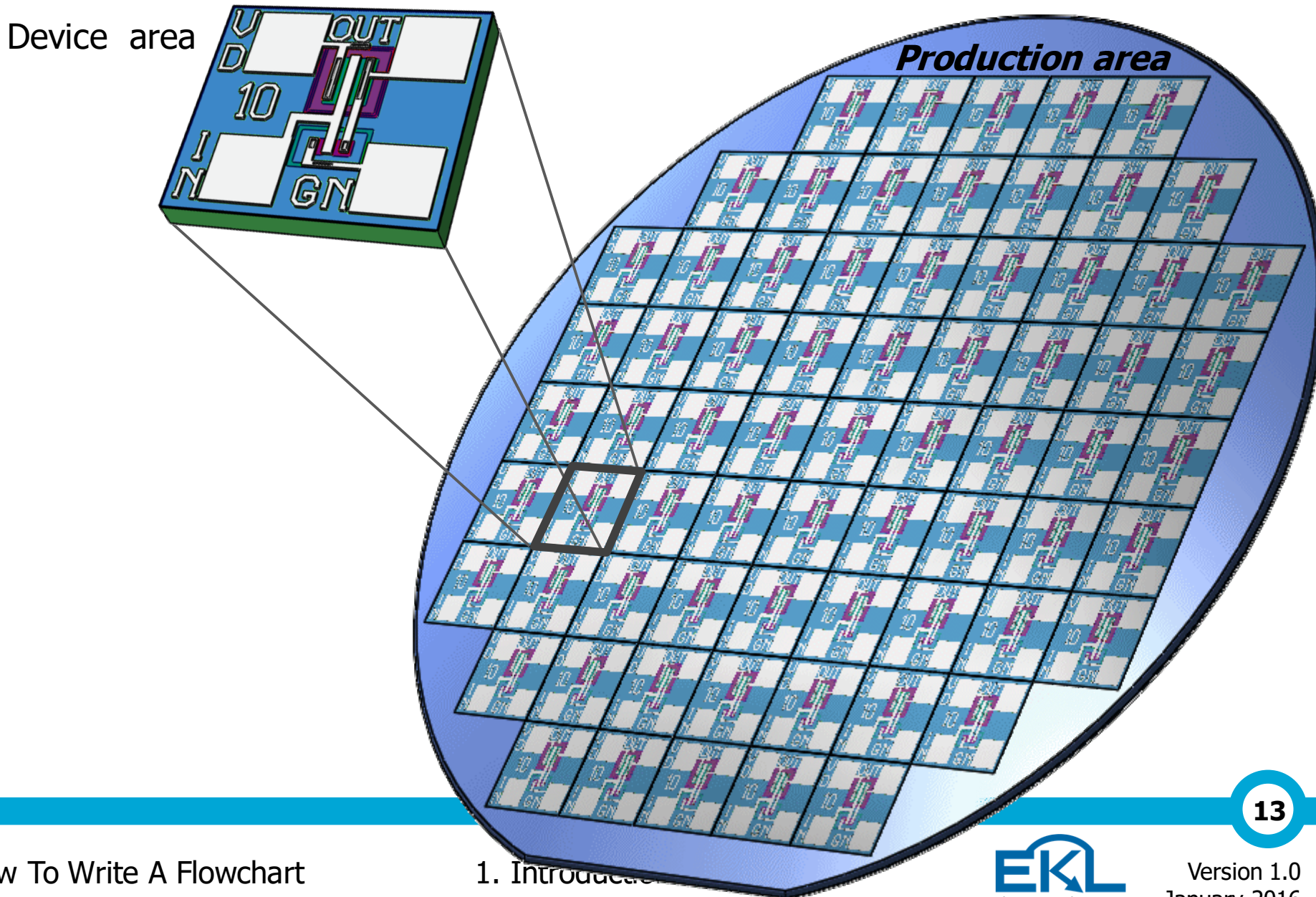
Small devices on a large production area.

The planar surface of a silicon wafer is processed in a sequence of process steps that result in a large number of devices on the wafer surface.

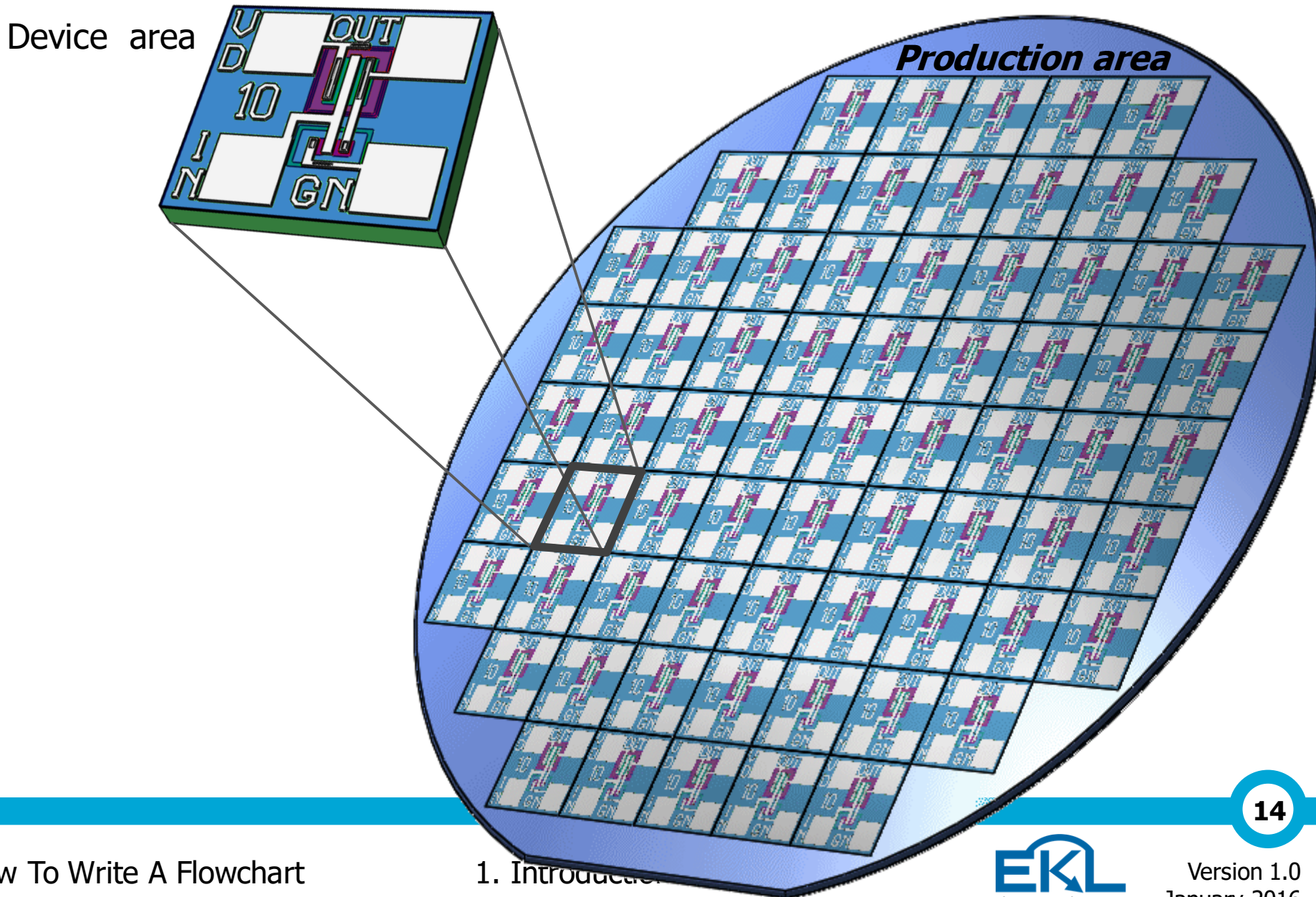
Production area



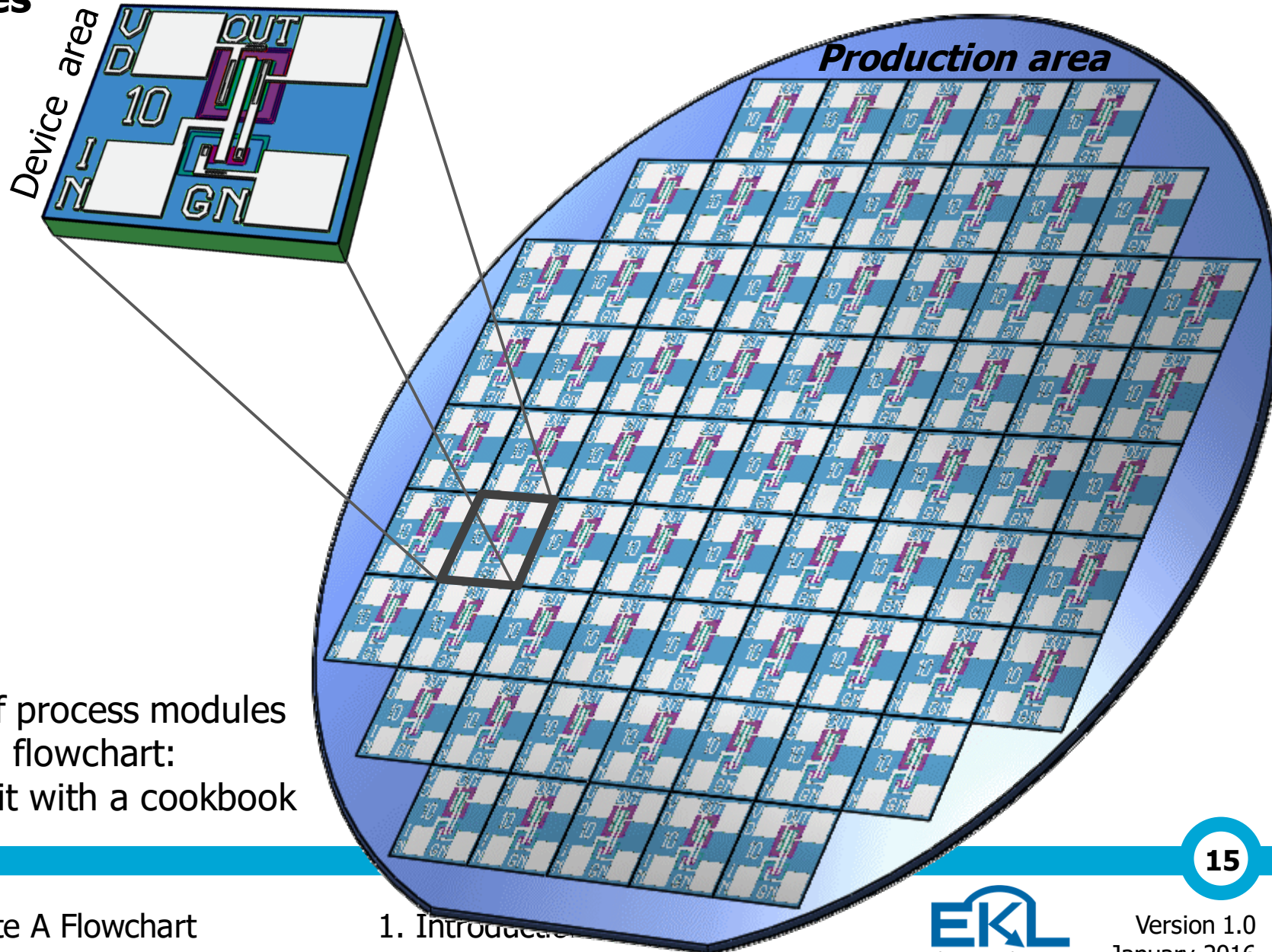
IC fabrication characteristics; 1 - planar technology



IC fabrication characteristics; 2 – parallel processing



IC manufacturing: a precisely defined sequence of process modules



The set of process modules is called a flowchart:
Compare it with a cookbook

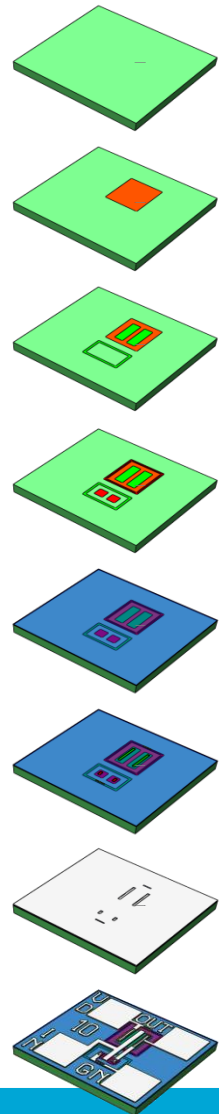
Process flow

How to cook an integrated circuit

STARTING MATERIAL.....	4
1. COATING AND BAKING wafer 1-2	4
2. ALIGNMENT AND EXPOSURE: Alignment makers wafer 1-2.....	4
3. WRITE WAFER ID wafer 1-2.....	4
4. DEVELOPMENT wafer 1-2.....	4
5. INSPECTION: LINEWIDTH AND OVERLAY wafer 1-2.....	4
6. DRY ETCHING: URK_NPD wafer 1-2.....	5
7. RHOTORESIST REMOVAL: Tepla wafer 1-2.....	5
8. CLEANING PROCEDURE: HNO3 100% and 65% (Si) wafer 1-2.....	5
9. DRY OXIDATION: DIRT BARRIER wafer 1-2.....	5
10. MEASUREMENT: OXIDE THICKNESS wafer 1-2.....	5
11. COATING AND BAKING wafer 1-2.....	5
12. ALIGNMENT AND EXPOSURE wafer 1-2.....	6
13. DEVELOPMENT wafer 1-2.....	6
14. INSPECTION: LINEWIDTH AND OVERLAY wafer 1-2.....	6
15. Phosphorus IMPLANTATION wafer 1-2.....	6
16. Phosphorus IMPLANTATION wafer 2.....	6
17. CLEANING PROCEDURE: Tepla + HNO3 100% and 65% (Si) wafer 1-2.....	6
18. ANNEALING wafer 1-2.....	6
19. MEASUREMENT: OXIDE THICKNESS wafer 1-2.....	7
20. OXIDE STRIP wafer 1-2.....	7
21. CLEANING PROCEDURE: Tepla + HNO3 100% and 65% (Si) wafer 1-2.....	7
22. DRY OXIDATION: DIRT BARRIER wafer 1-2.....	7
23. MEASUREMENT: OXIDE THICKNESS wafer 1-2.....	8
24. COATING AND BAKING wafer 1-2.....	8
25. ALIGNMENT AND EXPOSURE wafer 1-2.....	8
26. DEVELOPMENT wafer 1-2.....	8
27. INSPECTION: LINEWIDTH AND OVERLAY wafer 1-2.....	8
28. ARSENIC IMPLANTATION: NPN-EMITTER AND N-MOS SOURCE/DRAIN wafer 1-2.....	8
29. CLEANING PROCEDURE: Tepla + HNO3 100% and 65% (Si) wafer 1-2.....	9
30. COATING AND BAKING wafer 1-2.....	9
31. ALIGNMENT AND EXPOSURE wafer 1-2.....	9
32. DEVELOPMENT wafer 1-2.....	9
33. INSPECTION: LINEWIDTH AND OVERLAY wafer 1-2.....	9
34. BORON IMPLANTATION: INTRINSIC BASE AND PMOS SOURCE/DRAIN EXTENSIONS wafer 1-2.....	9
35. CLEANING PROCEDURE: Tepla + HNO3 100% and 65% (Si) wafer 1-2.....	9
36. COATING AND BAKING wafer 1-2.....	10
37. ALIGNMENT AND EXPOSURE wafer 1-2.....	10
38. DEVELOPMENT wafer 1-2.....	10
39. BORON IMPLANTATION: V-ADJUST_BOTTOM HALF wafer 1-2.....	10
40. CLEANING PROCEDURE: Tepla + HNO3 100% and 65% (Si) wafer 1-2.....	10
41. COATING AND BAKING wafer 1-2.....	10
42. ALIGNMENT AND EXPOSURE wafer 1-2.....	10
43. DEVELOPMENT wafer 1-2.....	10
44. BORON IMPLANTATION: V-ADJUST_RIGHT HALF wafer 1-2.....	11
45. CLEANING PROCEDURE: Tepla + HNO3 100% and 65% (Si) wafer 1-2.....	11
46. ANNEAL / OXIDATION wafer 1-2.....	11
47. MEASUREMENT: OXIDE THICKNESS wafer 1-2.....	11
48. COATING AND BAKING wafer 1-2.....	11
49. ALIGNMENT AND EXPOSURE: OXIDE THICKNESS adjust wafer 2.....	11
50. DEVELOPMENT wafer 1-2.....	12
51. INSPECTION: LINEWIDTH AND OVERLAY wafer 2.....	12
52. WET ETCHING SiO2 wafer 2.....	12
53. CLEANING PROCEDURE: Acetone + HNO3 100% (Si) wafer 2.....	12
54. COATING AND BAKING wafer 1-2.....	12
55. ALIGNMENT AND EXPOSURE CO wafer 1-2.....	12
56. DEVELOPMENT wafer 1-2.....	12
57. INSPECTION: LINEWIDTH AND OVERLAY wafer 1-2.....	13
58. WINDOW ETCHING wafer 1-2.....	13
59. CLEANING PROCEDURE: Acetone + HNO3 100% and HNO3 65% (Si) wafer 1-2.....	13
60. MARANGONI wafer 1-2.....	13
61. FIRST METALLIZATION: SPUTTERING OF Al/Si wafer 1-2.....	13
62. COATING AND BAKING wafer 1-2.....	14
63. ALIGNMENT AND EXPOSURE-IC wafer 1-2.....	14
64. DEVELOPMENT wafer 1-2.....	14
65. INSPECTION: LINEWIDTH AND OVERLAY wafer 1-2.....	14
66. WET ALUMINIUM ETCHING: 1 ST METAL wafer 1.....	14
67. POLY SILICON DIP ETCHING wafer 1.....	14
68. PLASMA ETCH ALUMINIUM: 1 ST METAL wafer 2.....	15
69. CLEANING PROCEDURE (after dry etching): Tepla wafer 2.....	15
70. CLEANING PROCEDURE (after wet etching): Acetone wafer 1.....	15
71. CLEANING PROCEDURE: HNO3 100% metal wafer 1-2.....	15
72. ALLOYING wafer 1-2.....	15
73. ELECTRICAL CHARACTERIZATION wafer 1-2.....	15

Vietnamese Home Cooking

Charles Phan
Chef-Owner
Tommy Vietina Food



How to write a flowchart

1. Introduction
- 2. Flowchart elements: process blocks, modules and steps**
3. Making a flowchart



Flowchart Hierarchy: Process blocks, modules and steps

Definition process flowchart:

A process flowchart is a connected sequence of process blocks, process modules and process steps to fabricate a device

Definition process block:

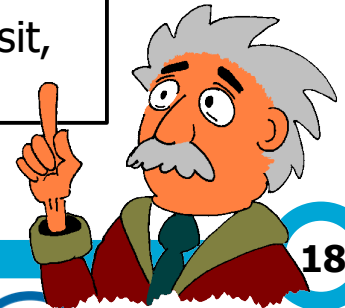
A process block is a connected sequence of process module and process steps which helps you to get the desired process target

Definition process Module:

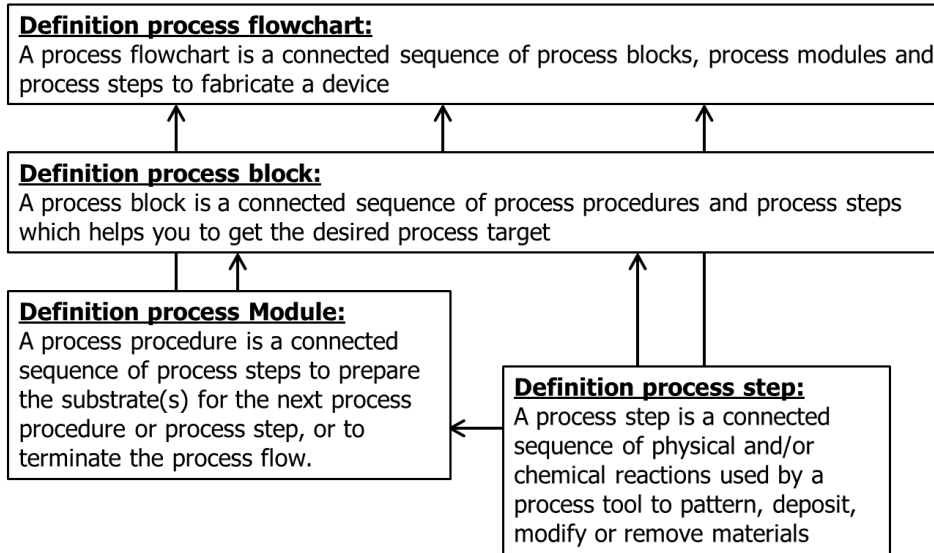
A process procedure is a connected sequence of process steps to prepare the substrate(s) for the next process procedure or process step, or to terminate the process flow.

Definition process step:

A process step is a connected sequence of physical and/or chemical reactions used by a process tool to pattern, deposit, modify or remove materials



Flowchart Hierarchy: Process blocks, modules and steps



Device level

Device component level

Tool level

Physical / chemical process level



19

IC process flowcharts: example of a process block

Flowchart

Process block

Process module

Process steps

Example:

Process block NW

8. CLEANING : HNO3 100% and 65% (Si)
9. DRY OXIDATION: DIRT BARRIER
10. MEASUREMENT: OXIDE THICKNESS
11. COATING AND BAKING
12. ALIGNMENT AND EXPOSURE-NW
13. DEVELOPMENT
14. INSPECTION: LINEWIDTH AND OVERLAY
15. Phosporus IMPLANTATION: N-WELL
16. CLEANING : Tepla + HNO3 100% and 65% (Si)
17. ANNEALING/NW-WELL DRIVE IN
18. MEASUREMENT: OXIDE THICKNES
19. OXIDE STRIPPING

IC process flowcharts: example of a process block

Flowchart

Process block

Process module

Process steps

Example:

Process block NW

8. CLEANING : HNO3 100% and 65% (Si)
9. DRY OXIDATION: DIRT BARRIER
10. MEASUREMENT: OXIDE THICKNESS
11. COATING AND BAKING
12. ALIGNMENT AND EXPOSURE-NW
13. DEVELOPMENT
14. INSPECTION: LINEWIDTH AND OVERLAY
15. Phosphorus IMPLANTATION: N-WELL
16. CLEANING : Tepla + HNO3 100% and 65% (Si)
17. ANNEALING/NW-WELL DRIVE IN
18. MEASUREMENT: OXIDE THICKNES
19. OXIDE STRIPPING

IC process flowcharts: example of a process module

Flowchart

Process block

Process module

Process steps

Example:

Process block NW

8. CLEANING : HNO₃ 100% and 65% (Si)
9. DRY OXIDATION: DIRT BARRIER
10. MEASUREMENT: OXIDE THICKNESS
11. COATING AND BAKING
12. ALIGNMENT AND EXPOSURE-NW
13. DEVELOPMENT
14. INSPECTION: LINEWIDTH AND OVERLAY
15. Phosphorus IMPLANTATION: N-WELL
16. CLEANING : Tepla + HNO₃ 100% and 65% (Si)
17. ANNEALING/NW-WELL DRIVE IN
18. MEASUREMENT: OXIDE THICKNESS
19. OXIDE STRIPPING

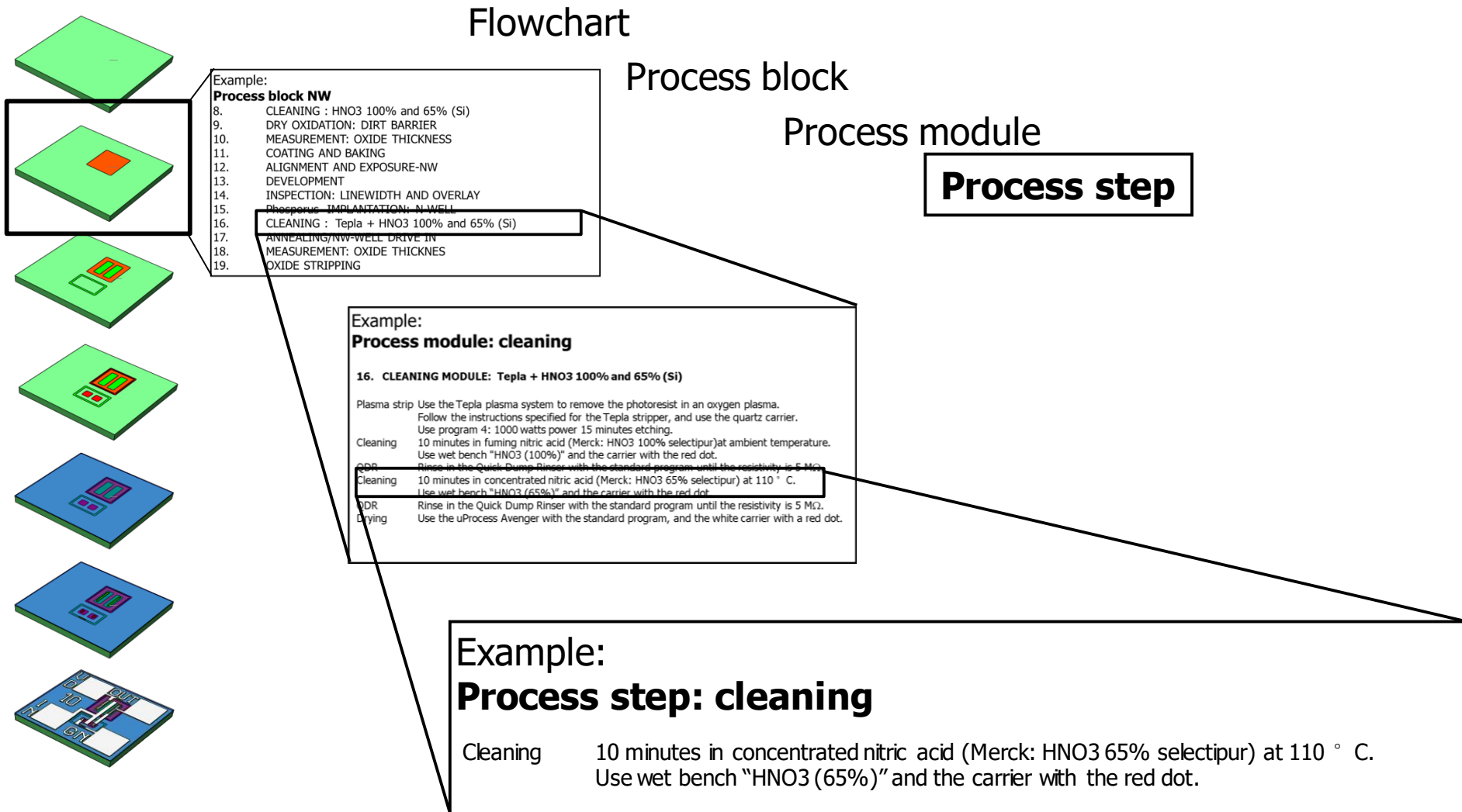
Example:

Process module: cleaning

16. CLEANING MODULE: Tepla + HNO₃ 100% and 65% (Si)

Plasma strip	Use the Tepla plasma system to remove the photoresist in an oxygen plasma. Follow the instructions specified for the Tepla stripper, and use the quartz carrier. Use program 4: 1000 watts power 15 minutes etching.
Cleaning	10 minutes in fuming nitric acid (Merck: HNO ₃ 100% selectipur) at ambient temperature. Use wet bench "HNO ₃ (100%)" and the carrier with the red dot.
QDR	Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 M Ω .
Cleaning	10 minutes in concentrated nitric acid (Merck: HNO ₃ 65% selectipur) at 110 ° C. Use wet bench "HNO ₃ (65%)" and the carrier with the red dot.
QDR	Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 M Ω .
Drying	Use the uProcess Avenger with the standard program, and the white carrier with a red dot.

IC process flowcharts: example of a process step



IC process flowcharts: example of a process step

Flowchart

Process block

Process module

Process step

Example:

Process block NW

8. CLEANING : HNO₃ 100% and 65% (Si)
9. DRY OXIDATION: DIRT BARRIER
10. MEASUREMENT: OXIDE THICKNESS
11. COATING AND BAKING
12. ALIGNMENT AND EXPOSURE-NW
13. DEVELOPMENT
14. INSPECTION: LINEWIDTH AND OVERLAY
15. Phosphorus IMPLANTATION: N-WELL
16. CLEANING : Tmp1 : HNO₃ 100% and 65% (Si)
17. ANNEALING/NW-WELL DRIVE IN
18. MEASUREMENT: OXIDE THICKNESS
19. OXIDE STRIPPING

Example:

Process step: annealing/NW-Well drive in

Furnace no:
Total time:

A1
415 min

Program name: OA002
Program no: N5

PROCESS	TEMPERATURE (in °C)	GASSES & FLOWS (in liter/min)	TIME (in minutes)	REMARKS
boat in	600	nitrogen: 3.0	5	
stabilize	600	nitrogen: 3.0	10	
heat up	+10 °C/min	nitrogen: 3.0 oxygen: 0.3	55	
stabilize	1150	nitrogen: 3.0 oxygen: 0.3	10	
drive-in	1150	nitrogen: 3.0 oxygen: 0.3	240	
cool down	-5 °C/min	nitrogen: 3.0	90	wait for operator
boat out	600	nitrogen: 3.0	5	



How to write a flowchart

1. Introduction
2. Flowchart elements: process blocks, modules and steps
- 3. Making a flowchart**



Flowchart ethics

Use a process run number.

- Process run number of you flowchart and relevant information on the first page
- Process run number on a label on your box
- Part of the file name of process related data files like documents, photo's, emails etc.

Make sure you get a process run number from Gregory Pandraud

Use EKL Label with your run number on your the wafer box.

Process work in the labs should be done with a hard copy of you flowchart.

- It is **the** source of information when you have any process related questions / discussion or issues with your process in the lab.
- It is the best location to note remarks to the flowchart, manual measurement data or other observations.

Flowchart Tips

Use a text editor with paragraph overview to edit your flowchart

- Each process step is listed as one item in the paragraph overview
- You can easily navigate through your flowchart
- You can easily modify flowchart steps

Print your flowchart on A4 cleanroom paper single sided.

- Use staples to make a book with the right page as blank to make notes/remarks related to the steps on the left page
- Include blank tables to note measurement results
- Write the file location of measurement data or photos in your flowchart
- Once finished, you can scan your flowchart to an electronic format and store for later reference. All the measurement data and notes are in relation to the flowchart steps.

Example: fabricate a metal thin film resistor on a silicon wafer

Basic equation of a thin film resistor:

$$R = \frac{\rho}{th} \cdot \frac{L}{W}$$

Where:

ρ = the resistivity of the conductive material

th = film thickness

L = length of the resistor

W = width of the resistor

What do we need?

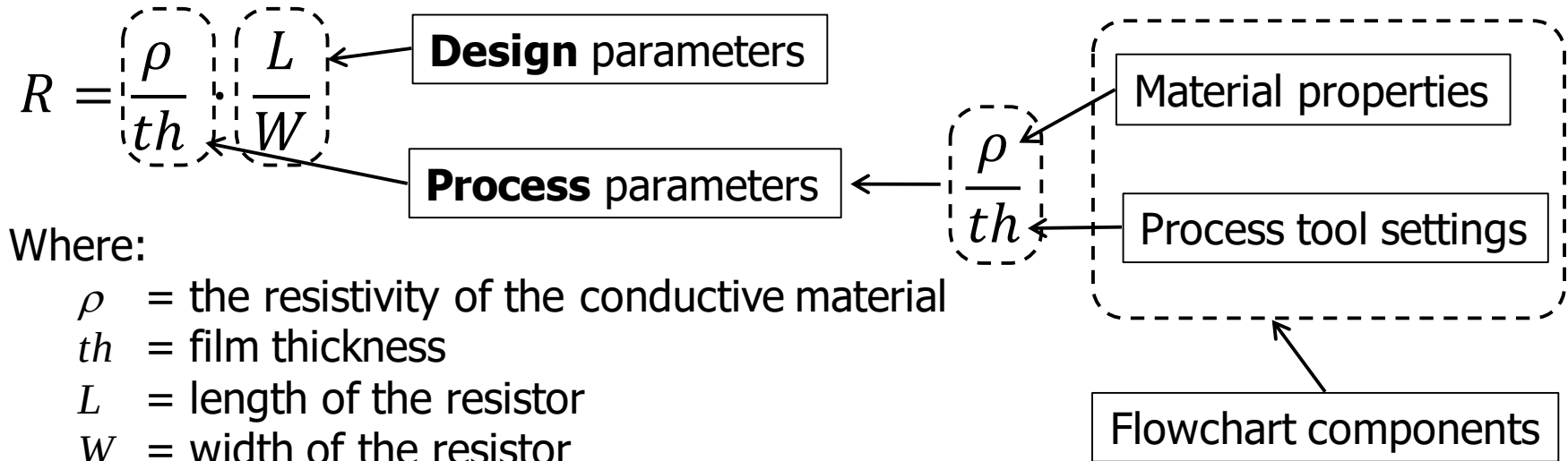
Electrical isolation of the surface

On top of that: a thin conductive film (ρ, th)

Pattern the thin film into a geometry (L, W) to obtain the right value for R

Example: fabricate a metal thin film resistor on a silicon wafer

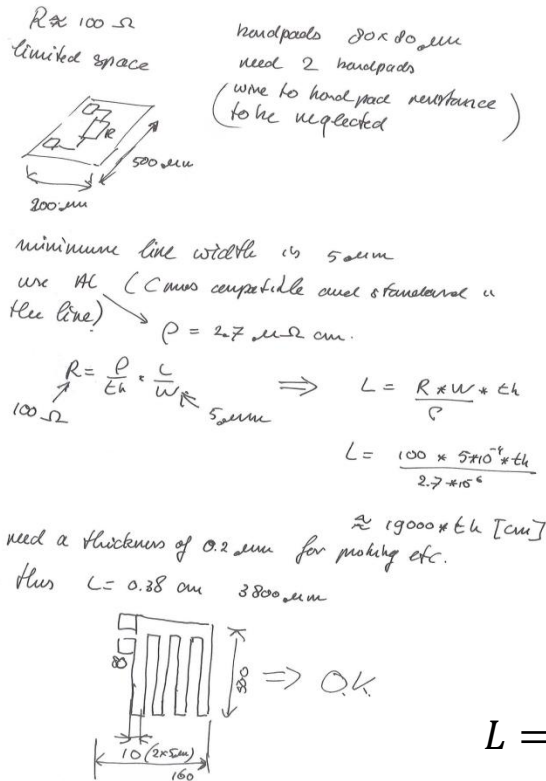
Basic equation of a thin film resistor:



The final result of a process = R

Which depends on **design** and **processing**

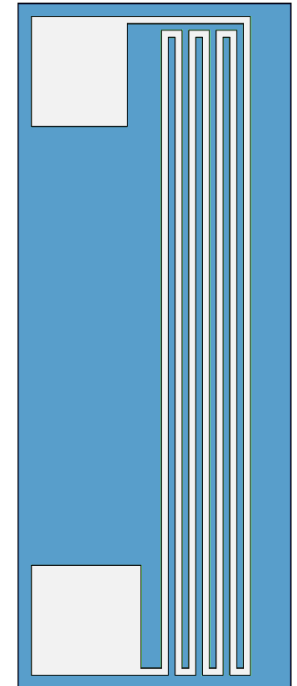
Draft a layout



Boundary conditions:

- $R = \frac{\rho}{th} \cdot \frac{L}{W} = 100 \Omega$
- Max. footprint = $200 \times 500 \mu m$
- Two bondpads of $80 \times 80 \mu m$ each (neglect the bondwire/bondpad resistance)
- CMOS compatible thin film resistor:
-> material of choice is AlSi(1%)
- Estimated minimum reliable thickness for wirebonding is 200 nm
- Minimum line width is $5 \mu m$

$$L = \frac{R \cdot W \cdot th}{\rho} = \frac{100 \cdot 5 \cdot 10^{-4} \cdot 0.2 \cdot 10^{-4}}{2.7 \cdot 10^{-6}} [\text{cm}] \rightarrow L = 0.38 \text{ cm}$$



Visualise the process flow in cross sections

Note that cartoon are applied to emphasize specific characteristics, therefore cross sections are usually **NOT** drawn to scale



Dhr. Rutte, prime
minister of the Netherlands



Visualise the process flow in cross sections



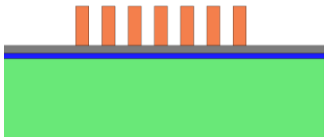
Start material, specify the required substrate



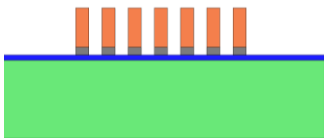
Thermal oxidation of silicon



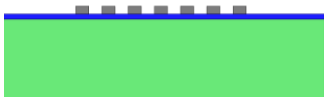
Deposit AlSi(1%) by sputtering



Photolithography: coat resist, expose and develop



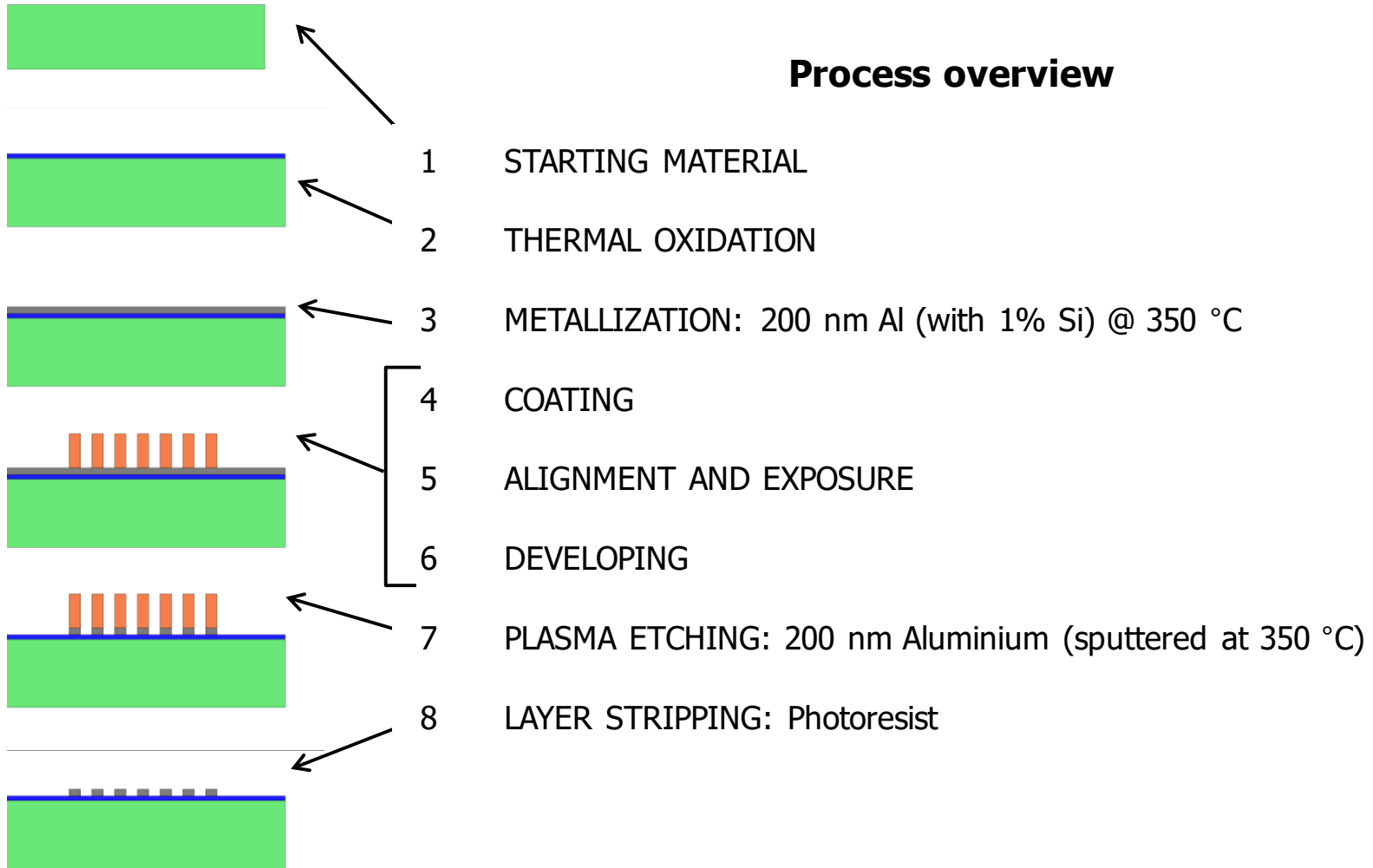
Etch the AlSi(1%)



Remove the photoresist

Translate the process flow in cross sections to process modules

Process overview



The use of platform flowchart as a template

What is a platform flowchart?

- A platform flowchart describes a proven process flow; a process flow that is processed before.

Why using a platform flowchart?

- This sequence of process steps does not lead to process conflicts and /or contamination issues.
- A valid starting point for your own flowchart.

How to use a platform flowchart?

- Identify a flowchart that match with the device or experiment you want to make.
- Copy that flowchart to a file with your run number in the file name.
- Remove unnecessary process steps without changing the sequence of the steps.
- Ask your mentor for a review.

From platform flowchart to customised flowchart

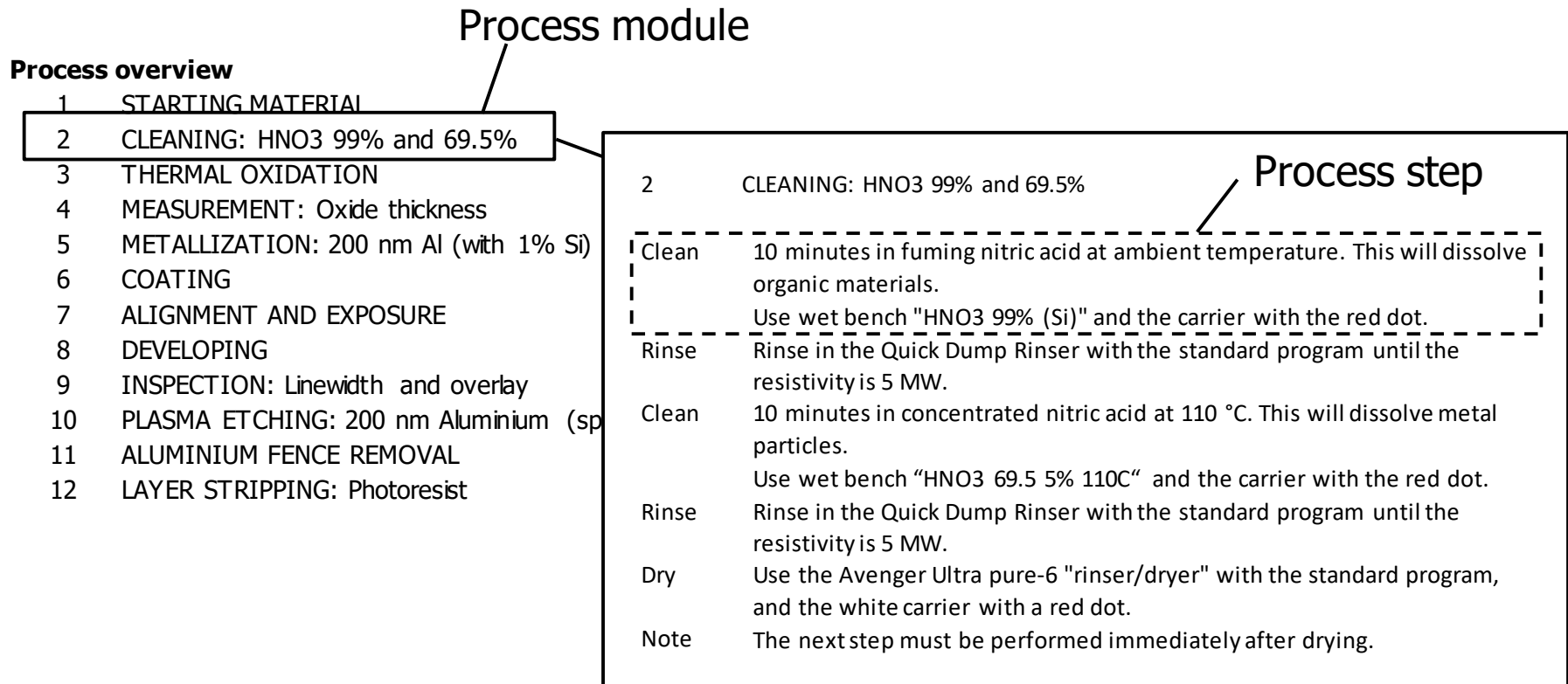
1. Start material-wafers

Brand name:	Topsil
Type:	DSP (p-type, boron)
Orientation:	<100>
Resistivity:	>1000 $\cdot$ cm
Thickness:	295-305 μ m
Diameter:	100 mm

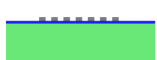
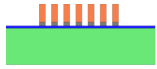
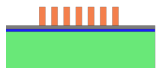
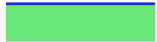
2. Non standard material: Copper, gold

3. Labs: CR 100; MEMS

From platform flowchart to customised flowchart



Feedback from your supervisor/mentor



Process overview

- 1 STARTING MATERIAL
- 2 THERMAL OXIDATION
- 3 METALLIZATION: 200 nm Al (with 1% Si) @ 350 °C
- 4 COATING
- 5 ALIGNMENT AND EXPOSURE
- 6 DEVELOPING
- 7 PLASMA ETCHING: 200 nm Aluminium (sputtered at 350 °C)
- 8 LAYER STRIPPING: Photoresist

Extra process blocks added, based on experience and way of working,....

Process overview

- 1 STARTING MATERIAL
- 2 CLEANING: HNO₃ 99% and 69.5%
- 3 THERMAL OXIDATION
- 4 MEASUREMENT: Oxide thickness
- 5 METALLIZATION: 200 nm Al (with 1% Si) @ 350 °C
- 6 COATING
- 7 ALIGNMENT AND EXPOSURE
- 8 DEVELOPING
- 9 INSPECTION: Linewidth and overlay
- 10 PLASMA ETCHING: 200 nm Aluminium (sputtered at 350 °C)
- 11 ALUMINIUM FENCE REMOVAL
- 12 LAYER STRIPPING: Photoresist

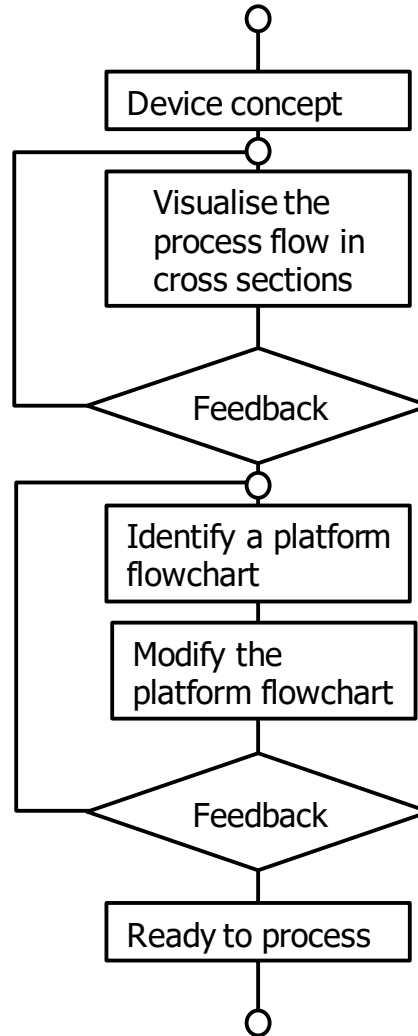
The whole flowchart is reviewed with your supervisor/mentor, special attention is given to avoid cross contamination

12. **UNR STOP RING Release kit**
Strip resist

Use the Tegra Plasma 3.00 system to remove the gold wire from an oxygen plasma. Follow the instructions specified for the Tegra stop gap, as discussed in the separate section.
Use a program 3 : 3.00 watts / min over an additional 10 min. (in this case 10 min. is not needed).

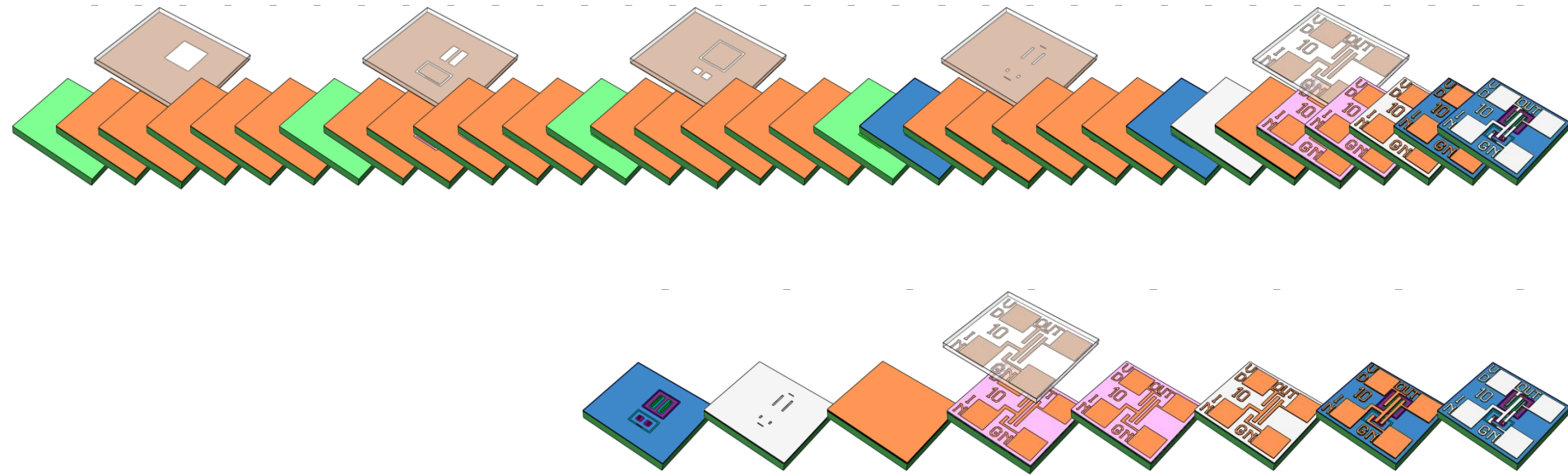


Flowchart of how to make a flowchart



Q&A

Flowchart Hierarchy: Process blocks, modules and steps



This slide will be used to visualise blocks/modules/steps hierarchy in a later version

Flowchart are not static

What to do in the lab, your flowchart can alter while working with it
Consult your supervisor/mentor

Process overview

- 1 STARTING MATERIAL
- 2 CLEANING: HNO₃ 99% and 69.5%
- 3 MARKER OXIDATION
- 4 MEASUREMENT: Oxide thickness
- 5 METALLIZATION: 200 nm Al (with 1% Si) @ 350 °C

- 6 CLEANING: HNO₃ 99% metal
- 7 COATING
- 8 ALIGNMENT AND EXPOSURE
- 9 DEVELOPING
- 10 INSPECTION: Linewidth and overlay
- 11 PLASMA ETCHING: 675 nm Aluminium (sputtered at 350 °C)
- 12 ALUMINIUM FENCE REMOVAL
- 13 LAYER STRIPPING: Photoresist

Time between metallisation and coating more that 24 hours, an intermediate cleaning step is required

